

A COMPARATIVE STUDY OF ARIMA AND LSTM MODELS TO FORECAST WALMART WEEKLY SALES

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ABSTRACT: This paper explores the application of time series analysis techniques, specifically ARIMA (Autoregressive Integrated Moving Average) and LSTM (Long Short-Term Memory), for predicting sales data. Accurate sales forecasting is essential for effective inventory management and strategic planning across various industries. By analyzing historical sales patterns, the research identifies underlying trends and variations. The ARIMA and LSTM models are then implemented to forecast future sales, with a focus on evaluating their performance in terms of accuracy and reliability. This study contributes valuable insights into the effectiveness of time series methods in enhancing sales forecasting accuracy, providing a framework for businesses seeking to optimize inventory management and improve decision-making processes.

Problem Statement: This paper outlines a methodology consisting of several steps to identify the most appropriate approach for demand forecasting in time series prediction. A summary of the process is provided in the next subsection.

INTRODUCTION: Predictive modelling plays a vital role in forecasting trends and supporting data-driven decisions across numerous fields. In finance, it aids in detecting fraudulent activities, while in healthcare, it helps evaluate the effects of environmental factors on public health. This study focuses on Walmart sales data to build predictive models using time series analysis, incorporating techniques such as autocorrelation, decomposition, and deep learning LSTM (Long Short-Term Memory). We concentrate on forecasting Walmart sales by preprocessing data, uncovering seasonal patterns and trends, and comparing model performance against historical records. The aim is to demonstrate how time series analysis can improve retail sales forecasting and support decision-making in a Walmart context.

LITERATURE REVIEW: The study found that LSTM outperformed ARIMA and SARIMA for profit prediction on large, complex datasets, using 46 years of sales records to compare model performance in a robust manner [1]. [3] highlights Focusing on fewer SKUs can optimize operations in a pull-based replenishment system. [2] evaluates ARIMA and SARIMA models for forecasting

COVID-19 cases and deaths using historical data. [4] findings aim to enhance understanding of the effectiveness of these models in predicting market movements despite data volatility. [7] finds that ARIMA and SARIMA models effectively forecast COVID-19 cases and deaths, with SARIMA being slightly better.

Data Collection: This paper examines the use of Walmart sales data from Kaggle to build predictive models using time series analysis. The dataset, collected from various Walmart stores in the U.S. between February 5, 2010, and October 26, 2012, includes weekly sales data along with features like holiday flags, temperature, fuel prices, CPI, and unemployment rates.

Outlier detection:

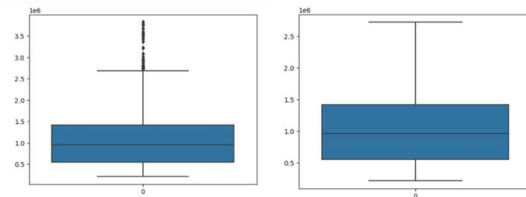


Figure 1: Identified possible outliers in the Weekly Sales column

Decomposition plot: A decomposition plot helps reveal the main structure in Walmart sales data by showing key trends, patterns, and seasonal changes. This insight supports better forecasting and decision-making.

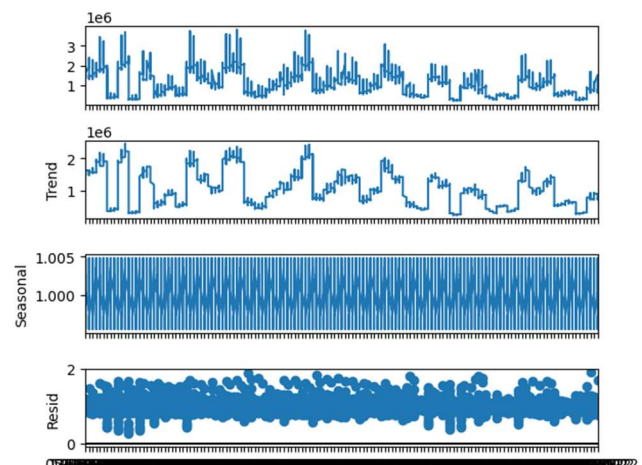


Figure 2: Seasonal Variance of weekly sales

Seasonal Variation: Periodic events like holidays, the back-to-school season, summer breaks, and winter holidays greatly influence consumer spending habits and preferences.

Impact on Sales: This recurring pattern highlights the effect of seasonal factors on Walmart's sales outcomes.

Lagging the data: ACF and PACF plots are frequently used in time series analysis to analyze the autocorrelation patterns in data over time. These plots assist in determining suitable AR and MA components by showing how the data relates to itself at different time lags.

ACF Plot:

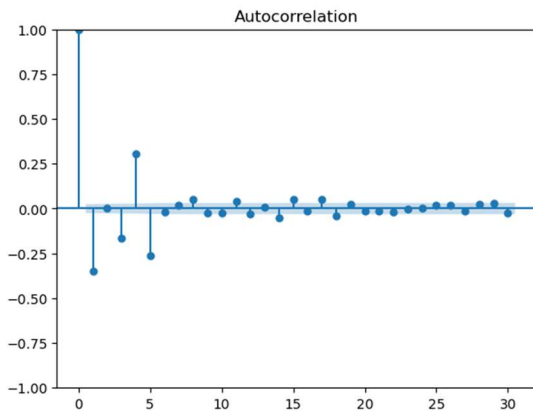


Figure 3: ACF plot of outlier removed data

PACF Plot:

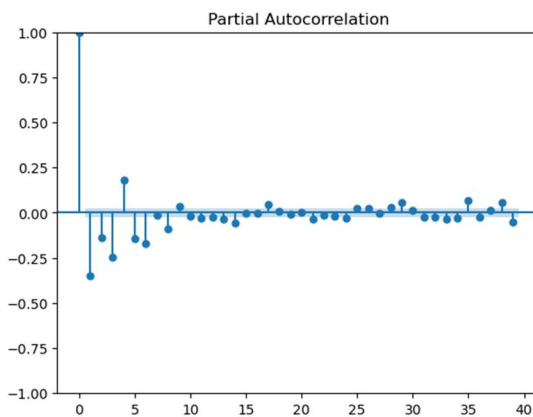


Figure 4: PACF plot of outlier removed data

Frameworks used:

Pmdarima: Automatically picks the best settings for ARIMA.

TensorFlow and **Keras** are popular frameworks for creating and training neural network models in deep learning.

ARIMA: To analyze Walmart's weekly sales using an ARIMA model, first, choose three key parameters: p (the number of past sales data points to consider) d (the number of times to difference the data for stationarity), and q (the

size of the moving average to correct past prediction errors). After selecting these parameters, fit the ARIMA model using a library like Statsmodels in Python with the command `model = ARIMA(data, order=(p, d, q))`. Next, check the model fit by examining the residuals, which should appear random without clear patterns. Finally, use the fitted model to forecast future weekly sales, enabling Walmart to better understand sales trends and make informed decisions.

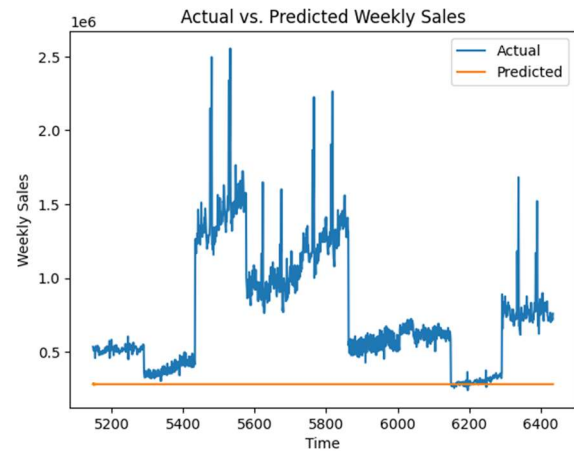


Figure 5: Prediction using ARIMA

LSTM: To analyze Walmart's weekly sales using an LSTM model, begin by preparing your data, ensuring it's in a suitable format for time series forecasting. Normalize the sales data to improve model performance, and then split it into training and testing datasets. Next, structure the data into sequences, where each sequence consists of past sales data used to predict future sales. Build the LSTM model using a deep learning framework like TensorFlow or Keras, specifying the number of layers and units. Train the model on the training data, adjusting parameters like batch size and number of epochs for optimal performance. After training, evaluate the model's accuracy on the testing set and make predictions for future weekly sales. This approach enables Walmart to leverage deep learning techniques for capturing complex patterns and trends in sales data.

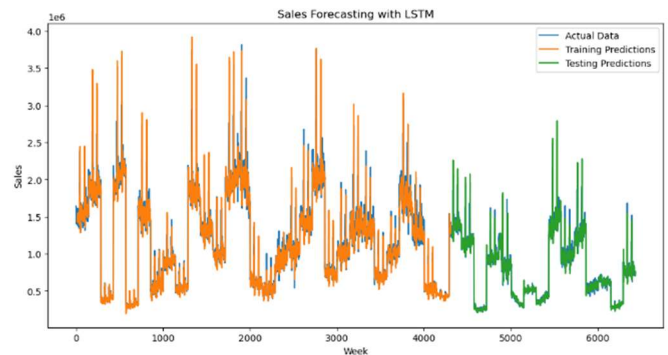


Figure 6: Prediction using LSTM

OUTLINE:

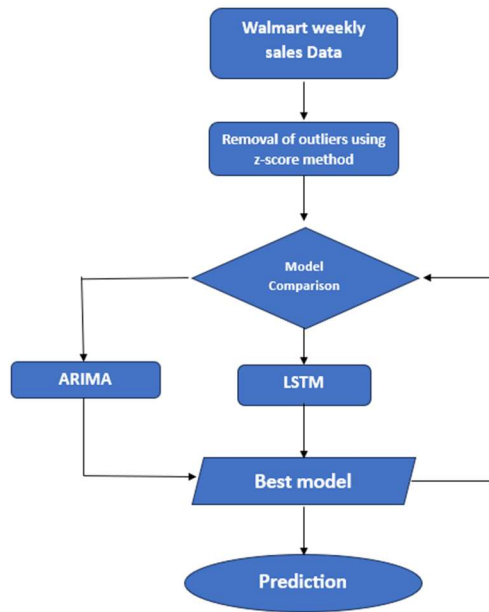


Figure 7: Methodology Flow chart of the paper

PERFORMANCE METRICS:

	ARIMA	LSTM
RMSE	617840.12360	1295970.03
MAE	480422.1446	0.0237872
MAPE	52.6244	0.00190741

Table 1: Comparison of performance metrics

CONCLUSION: Time series forecasting of the retail sales is a difficult endeavour that demands complex methodologies and meticulous study. LSTM model and ARIMA time series forecasting model have been routinely utilized to forecast retail market values. In the above comparison table, the LSTM model has much lower MAE and MAPE values, indicating a more accurate prediction on average compared to ARIMA. The RMSE for LSTM is higher, but because MAE and MAPE are generally better indicators of accuracy in prediction, LSTM still appears to be the better model overall in this context. Finally, we may infer that the LSTM model fits the presented data better than the ARIMA model based on the assumption of figure 5 and figure 6.

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